# Key Insights & Recommendations

**One thing to know before reading:** this dataset has no sales numbers. So when I say “inventory value” or “revenue”, I mean selling price × available quantity. It shows what is sitting on the shelf, not what was actually sold.

## Main Findings

1. **Cheap products go out of stock the most.** Products under ₹100 were out of stock 17.84% of the time. For ₹100 to ₹500 it was 6.96%, and for ₹500 and above it was only 1.79%.
2. **Moderate discounts go with better availability.** Products with a 10–40% discount rarely go out of stock (6.15% and 0.79%), while products with a 0–10% discount go out of stock 15.32% of the time.
3. **A few heavy discounts cost a lot.** Among the top 15 heavily discounted products, one olive oil alone accounts for about 38% of the total discount loss (₹7,206 of ₹18,780).
4. **Salt, onion, potato and atta give the least value per gram** among the heavy products (1 kg and above), so they are the hardest items to make worth delivering.

## Section A: Stock Availability

### What I found

- Biscuits has the highest stockout rate at 28.57% (42 of 147 products). Dairy, Bread & Batter and Beverages follow at 21.71% each, then Meats, Fish & Eggs at 19.05%.
- The lowest stockout rates are in Personal Care and Paan Corner (6.10%) and Fruits & Vegetables (6.45%).
- A lot of in-stock products are down to their last unit. Most of them are cheap, but not all. Ariel Matic Top Load detergent (₹499) and Bagrry's muesli (₹595) are also on the list.
- By discount band: 0–10% had 2,447 products and a 15.32% stockout rate, 10–25% had 1,074 and 6.15%, 25–40% had 127 and 0.79% (just 1 product), and 40%+ had 79 and 13.92%.

### What I would suggest

- Keep a bigger buffer for Biscuits, Dairy and Beverages.
- Set an automatic reorder alert at 1–2 units left, starting with higher-priced products.
- Test 10–25% promotions on products that often go out of stock. The 40%+ band has only 79 products, so treat its result with caution, and note that this shows a pattern, not proof that discounts cause better availability.

## Section B: Inventory Value & Warehouse Efficiency

### What I found

- Everything in stock adds up to about ₹22.4 lakh at selling price.
- Cooking Essentials and Munchies hold the most value (₹3.37 lakh each, about 15% of the total), followed by Personal Care and Paan Corner (₹2.71 lakh each). Fruits & Vegetables holds the least (₹10,846).
- No in-stock product has more than 6 units. The products with the largest stock are low-priced items (₹10–20, such as noodles, cookies and masala mixes) that hold the same 6 units as much pricier products.
- Out-of-stock products are not equal in value. Meats, Fish & Eggs has only 12 products out of stock, but their listed value is ₹2,024 (about ₹169 per product). Biscuits has 42 products out of stock worth ₹1,650.50 (about ₹39 per product).

### What I would suggest

- Check the reorder quantities in the highest-value categories first, because that is where most of the money is sitting.
- Set stock levels by price and demand instead of the same level for every product, so shelf space is not used up by ₹10–20 items.
- When restocking, look at the value missing and not only the number of products. Meats, Fish & Eggs has fewer gaps than Biscuits, but each gap costs more.

## Section C: Discount Strategy & Margin

### What I found

- Cooking Essentials and Munchies give away the most discount in rupees (₹41,252 each) even though their average discount is only 7.35%.
- Fruits & Vegetables has the highest average discount (15.79%) and Meats, Fish & Eggs is next (12.63%), but their total discount cost is small (₹2,071 and ₹4,023).
- Some products with 45–50% discount still have stock: Borges Extra Light Olive Oil (46%), Pampers Pants Large (45%), RRO cheeses, Moi Soi sauces and Ceres instant masalas (50%). Each has only 3 to 6 units, but because the MRP is high, the discount value adds up (₹7,206 for the olive oil and ₹2,100 for the Pampers).
- Staples like 24 Mantra organics, Aashirvaad atta and ghee, and Ariel detergents have 0% discount and 6 units in stock.
- Expensive products (MRP above ₹500) like Dhara, Saffola and Fortune oils, Surf Excel, Pedigree, Nestle Nan Pro and L'Oreal hair colour have 0–9% discount, and every one visible in my result is in stock.

### What I would suggest

- Look at how many rupees a discount costs, not only the percentage. Cooking Essentials and Munchies are the first place to review.
- Put a limit on discounts for high-value products like the olive oil, and use short-time or targeted offers instead.
- Try a small 3–5% discount, or a bundle (atta with ghee, for example), on the full-priced staples that have full stock.
- Keep premium products close to full price, and discount only the ones that sell slowly.

## Section D: Pricing & Delivery Efficiency

### What I found

- Products under ₹100 are 1,833 of the 3,727 listings (about 49%). ₹100–500 has 1,782 and ₹500 and above has only 112.
- Stockout rate drops as price goes up: 17.84% for Low, 6.96% for Mid and 1.79% for Premium.
- Among heavy products (1 kg and above), the lowest value per gram is in daily staples: salt (1 kg for ₹19–36), onion (3 kg for ₹57), potato (3 kg for ₹84) and atta (10 kg for ₹375–399).

### What I would suggest

- Pay more attention to restocking budget products. They are the biggest part of the catalog and the ones that run out most.
- A minimum order value, a delivery fee, or bundling these staples with higher-priced items could help cover the delivery cost of heavy, low-value products.

## Section E: Product Performance

### What I found

- The products with the highest estimated revenue are expensive items with full stock. In Cooking Essentials the top three are Borges Extra Light Olive Oil (₹8,394), Praakritik A2 Ghee (₹7,830) and Saffola Gold (₹7,440).
- Seven of the 14 categories hold about 84% of the total inventory value. Cooking Essentials and Munchies together hold about 30%, and the three smallest categories (Fruits & Vegetables, then Meats, Fish & Eggs, then Biscuits) hold only about 2.5% together.

### What I would suggest

- Make sure the top 3 products of each category never go out of stock.
- Spend most of the planning and monitoring effort on the highest-value categories.

## Limitations

- There is no sales data and no cost data, so everything here is based on listed prices and stock, not real sales or profit.
- Available quantity is small (no in-stock product has more than 6 units), so inventory value depends mostly on price and the number of listings.
- The results show patterns, not proof of cause, for example between discounts and stockouts.
- Some categories have exactly the same product counts and values (for example, Cooking Essentials and Munchies). This suggests the same products appear under more than one category in the source data. I kept the category results as they are, so category comparisons should be read with that in mind.